Typology-Clustered Latent GraphSMOTE & Parametric Bilinear Edge Generation

Step 1: Typology Clustering

- Minority illicit partition
- Cluster into $C_1, \ldots, C_K$
- Preserves distinct rings
- Prevents mode collapse

Step 2: Latent Manifold Interp.

- Sample $u, v \in C_k$ in train set
- $\mathbf{h}_{\text{syn}} = (1-\rho)\mathbf{h}_u + \rho\mathbf{h}_v$
- $\rho \sim \mathcal{U}(0, 1)$ on manifold
- Generates realistic mules

Step 3: Bilinear Link Gen.

- Link to real node $w \in \mathcal{V}_{\text{train}}$
- $\hat{\mathbf{A}}_{\text{syn},\,w} = \sigma(\mathbf{h}_{\text{syn}}^T \mathbf{W}_{\text{edge}} \mathbf{h}_w)$
- Threshold $\tau_{\text{edge}} = 0.60$
- Reconstructs flow graph

Step 4: Zero-Leakage Barrier

- 100% confined to D_{train}
- 0 edges to D_{cal} or D_{test}
- Standalone GNN recall: **10.33% → 90.10%** (+79.77 pp boost)